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Early life supplementation with a natural blend containing turmeric, thymol, and yeast cell wall components to optimize rumen anatomical and microbiological development and productivity in dairy goats

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ABSTRACT

Ruminants are born with an anatomically, microbiologically, and metabolically immature rumen. Optimizing the rearing of young ruminants represent an important challenge in intensive dairy farms. Therefore, the objective of this study was to evaluate the effects of dietary supplementation of young ruminants with a plant extract blend containing turmeric, thymol, and yeast cell wall components such as mannan oligosaccharides and β-glucans. One hundred newborn female goat kids were randomly allocated to 2 experimental treatments, which were unsupplemented (CTL) or supplemented with the blend containing plant extracts and yeast cell wall components (PEY). All animas were fed with milk replacer, concentrate feed, and oat hay, and were weaned at 8 wk of age. Dietary treatments lasted from wk 1 to 22 and 10 animals from each treatment were randomly selected to monitor feed intake, digestibility, and health-related indicators. These latter animals were euthanized at wk 22 of age to study the rumen anatomical, papillary, and microbiological development, whereas the remaining animals were monitored for reproductive performance and milk yield during the first lactation. Results indicated that PEY supplementation did not lead to feed intake or health issues because PEY animals tended to have a higher concentrate intake and lower diarrheal incidence than CTL animals. No differences between treatments were noted in terms of feed digestibility, rumen microbial protein synthesis, health-related metabolites, or blood cell counts. Supplementation with PEY promoted a higher rumen empty weight, and rumen relative proportion to the total digestive tract weight, than CTL animals. This was accompanied with a higher rumen papillary development in terms of papillae length and surface area in the cranial ventral and caudal ventral sacs, respectively. The PEY animals also had higher expression of the *MCT1* gene, which is related to volatile fatty acid absorption by the rumen epithelium, than CTL animals. The antimicrobial effects of the turmeric and thymol could explain the decreased the rumen absolute abundance of protozoa and anaerobic fungi. This antimicrobial modulation led to a change in the bacterial community structure, a decrease in the bacteria richness, and to the disappearance (i.e., *Prevotellaceae_UCG-004*, *Bacteroidetes_BD2–2*, *Papillibacter*, *Schwartzia*, and *Absconditabacteriales_SR1*) or decline of certain bacterial taxa (i.e., *Prevotellaceae_NK3B31_group*, and *Clostridia_UCG-014*). Supplementation with PEY also decreased the relative abundance of fibrolytic (i.e., *Fibrobacter succinogenes* and *Eubacterium ruminantium*) and increased amylolytic bacteria (*Selenomonas ruminantium*). Although these microbial changes were not accompanied with significant differences in the rumen fermentation, this supplementation led to increased body weight gain during the preweaning period, higher body weight during the postweaning period, and higher fertility rate during the first gestation. On the contrary, no residual effects of this nutritional intervention were noted on the milk yield and milk components during the first lactation. In conclusion, supplementation with this blend of plant extracts and yeast cell wall component in early life could be considered as a sustainable nutritional strategy to increase body weight gain and optimize the rumen anatomical and microbiological development in young ruminants, despite having minor productive implications later in life.

Key words: antimicrobial, curcumin, essential oils, rumen microbiota

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INTRODUCTION

The current societal demands on livestock production require using sustainable management systems to allow optimizing the productivity and health in young ruminants. The viability of any dairy farm rest, to a great extent, on a successful rearing program of young animals for replacement. Suboptimal managements in early life have been associated with postweaning shock in growth and productive and reproductive insufficiencies during the adulthood (Curtis et al., 2018). A smooth transition from milk to solid feed occurs when newborns are reared with the dam, mostly due to a natural rumen microbial transfer to the offspring (Belanche et al., 2019c) and feeding behavior learned from the dam (De Paula Vieira et al., 2012). Contrarily, in most intensive dairy systems, newborns are fed on milk replacer, or whole milk, with no contact with adult animals. This artificial rearing system has been shown to limit or delay the rumen microbiological (i.e., lower bacterial diversity and absence of protozoa) and fermentative activity, having negative effects on feed digestibility and productivity (Belanche et al., 2019c). These adverse effects can increase when artificial rearing is combined with early weaning programs (Lu and Potchoiba, 1988), and animals often suffer health problems and growth retardation, which often require antibiotic therapies during this critical period (McCoard et al., 2021).

In contrast to adult animals in which the rumen microbiota is relatively stable, young animals have a higher plasticity and the modification of the gut colonization pattern toward a desirable microbial community could potentially have long-lasting effects on productivity (Yáñez-Ruiz et al., 2015). Moreover, recent research has been focused on the identification of natural feed additives with the ability to simultaneously modulate the rumen microbiota (Belanche et al., 2016) and to promote a satisfactory rumen papillae development (Zhang et al., 2021a) because both processes are intimately linked (Diao et al., 2019). Therefore, novel early-life nutritional strategies should be developed to improve animal health and an optimal gut anatomical, microbiological, and functional development to ensure a satisfactory productivity in adult life.

Among the different nutritional strategies available, the use of plant extracts and probiotics has gained attention because of their antimicrobial properties (Calsamiglia et al., 2007). In particular, the thymol (5-methyl-2-isopropylphenol) is a natural essential oil (**EO**) derived of p-cymene, isomeric with carvacrol, and represents the main active component of several plants such as thyme (*Thymus vulgaris*) and oregano (*Origanum vulgare*). The use of thymol, or plants containing this active molecule, has sometimes been associated with increases in the ADG and feed utilization in dairy cattle (Wu et al., 2020a,b), and lower diarrhea rate in calves (Liu et al., 2020). Turmeric is the root of the herbaceous perennial plant *Curcuma longa* and has traditionally been used for medicinal and dietary purposes (Lans et al., 2007). Turmeric is rich in curcuminoids, such as curcumin (70–77%), demethoxycurcumin (18–20%), and bisdemethoxycurcumin (7–10%), which exhibit a wide range of medicinal properties (Tyagi et al., 2015). Turmeric has been used as feed additive due to its flavoring, and antimicrobial, antiinflammatory, and antioxidant properties (Lans et al., 2007; Khalesizadeh et al., 2011). Curcumin has shown positive rumen antimicrobial effects in vitro (Aderinboye and Olanipekun, 2021) and antioxidant and antiinflammatory effects in vivo resulting on higher milk yield in dairy sheep (Jaguezeski et al., 2018). Dietary supplementation with yeast cell wall (**YCW**) oligosaccharides could represent another nutritional alternative to antibiotics because it can improve the intestinal barrier, absorb pathogens, and enhance the release of cytokines, leading to improved immunity in broilers (Liu et al., 2018). In ruminants, several studies have demonstrated that YCW supplementation is beneficial for milk production (Nocek et al., 2011) and ruminal development (Lesmeister et al., 2004).

Despite the promising results on the use of different plant extracts and YCW components, the mechanisms that control the effects of these nutritional strategies are largely unknown and the potential synergetic effects when these types of additives are supplemented in combination require further research. The few attempts of combining these compounds on rumen fermentation, blood parameters, and productive performance have shown inconsistent results due to the use of different dosages, type of diets, and ruminant species (Biricik et al., 2016; Alemu et al., 2019). Furthermore, there is a lack of knowledge in relation to the potential short- and long-term effects of feeding these additives to young ruminants.

This study aims to optimize the artificial rearing of dairy goats through a nutritional strategy based on the combination of various active compounds (i.e., turmeric, thymol, and YCW). It was hypothesized that this novel nutritional approach could result in additional improvements in the rumen anatomical, microbiological, and physiological development, which could lead to short- and long-term positive effects. A holistic approach based on the study of the rumen microbiota, histology, gene expression, nutrient utilization, and productive outcomes was used to achieve this goal.

MATERIALS AND METHODS

Animals and Diets

Animal procedures were conducted according to the Spanish guidelines (RD 53/2013) and protocols were approved by the Ethical Committee of Animal Research (EEZ-CSIC) and regional government (ref. 27/03/2020/042). A total of 100 female goat kids born within a 2-wk period were used from an intensive commercial Murciano-Granadina dairy goat farm located in Bogarre, southern Spain (37°41′N, −3°36′W 1,090 m altitude) during summer 2019. All animals remained in the same building and were reared following a conventional farm management. Immediately after birth, kids were fed colostrum (ca. 250 mL) and randomly assigned to 2 experimental groups, which were initially balanced by BW. One group (PEY, n = 50) was dietary supplemented with a plant extract blend (supplied by CCPA Group) containing turmeric (0.13 g/kg of additive), thymol (1.88 g/kg of additive), and YCW components such as mannan oligosaccharides (75 g/kg of additive) and β-glucans (125 g/kg of additive) as the main active components, whereas the control group (CTL, n = 50) received no additives. These dosages were chosen based on the manufacturer's recommendations supported by a previous in vitro study (Macheboeuf et al., 2008). All animals were raised on milk replacer (declared composition in DM: 91.2% OM, 23.0% CP, 23.0% fat, and 1.1% Ca), which was offered ad libitum and freshly prepared by mixing 185 g/L of milk powder (Bacilactol Corderos y Cabritos Prima, Nuter Feed S.A.U.; Supplemental Table S1; https://figshare.com/articles/journalcontribution/2023JDS_Suppldocx/22707937; Belanche et al., 2023) in the same milk feeder. The dietary intervention was applied to the PEY group from d 2 (after receiving the colostrum) to wk 22 of age, and both groups were grouped afterward in the same herd. The format of the additive was adapted to the age of the animals to achieve the target doses of 0.03, 0.45, 18, and 30 mg/kg of BW for turmeric, thymol, mannan oligosaccharides, and β-glucans, respectively. From d 2 of age to weaning (wk 8), the feed additive was mixed with licking clay (5% inclusion rate in DM), whereas after weaning, it was mixed with the concentrate feed (0.7% inclusion rate in DM). The CTL group also had free access to the same licking clay and concentrate feed but without additive. From wk 2 of age, all animals had ad libitum access to a pelleted starter concentrate (SPG Starter for lambs and kids, ALIMER, UAG; Supplemental Table S1) with a 3-mm diameter (declared composition in DM: 94% OM, 18.0% CP, 4.08% fat, 3.20% crude fiber). All animals were weaned at 8 wk of age by progressively decreasing the concentration of milk replacer powder in the reconstituted milk during 7 d. After weaning, all animals had free access to cereal straw and the same concentrate feed. Body weight was monitored every 2 wk using a digital dynamometer (model CR-S, Gram) and a digital scale (model GE1–150-C, Campesa) prior and after weaning, respectively. Presence of diarrhea and its severity were monitored every 2 wk by the same trained person following a defined score (Bentounsi et al., 2012): 1 corresponding to normal feces in pellets, 2 corresponding to soft feces (similar to cow pat), 3 corresponding to semi-liquid feces, and 4 corresponding to profuse diarrhea with liquid feces. Ten females per group were sold at weaning and did not continue in the study.

Rumen Fermentation, Feed Utilization, and Animal Performance

At wk 19 of age, 10 females from each group were transported to the EEZ-CSIC Animal Research Service (Granada, Spain) and fed with the same experimental diets. After 3 wk of adaptation to the research facilities, animals were placed in individual pens (2 × 2 m) with free access to the commercial concentrate and oat hay. The incidence of diarrhea (fecal score from 1 to 5) was monitored weekly. Due to the occurrence of diarrhea at wk 19 of age, feces were sampled with swabs and a microbial culture was performed in Petri plates using Agar MacConkey Number 3 and Agar XLD (Oxoid PO5002A and PO5057A, respectively) in an external laboratory (Exopol, San Mateo de Gállego, Spain). Microbial cultures were aerobically incubated at 37°C for 48 h and colonies compatible with *Escherichia coli* were monitored using a semiquantitative score as follows: 1 corresponding to minor colony growth (1–10 cfu/swab), 2 to moderate growth (11–50 cfu/ swab), and 3 to massive growth (>50 cfu/swab). The *E. coli* cultures were confirmed by MS (MALDI-TOF, Bruker) based on the fingerprinting of the large biomolecules and using a *E. coli* strain (CECT 4972) as calibration standard as previously reported (Singhal et al., 2015).

Voluntary feed intake of concentrate and forage was daily monitored during 5 consecutive days in wk 21. The amount of diet offered (ad libitum) and fed residues were monitored from each individual animal. On the wk 22 of age, animals were placed on individual metabolic crates for 4 consecutive days to determine N balance and total-tract apparent digestibility of dietary nutrients. Total feces and urine production were daily monitored. Chemical composition in feeds and feces was performed following the AOAC International (2005) for DM (method 934.01), OM (method 942.05),

and N (method 990.03) using the Dumas method (Luco TruSec CN). Concentrations of NDF and ADF were measured using the Ankom 2020 fiber analyzer unit (Ankom Technology Corp.) and expressed without residual ash. The gross energy content was measured determined by using an adiabatic calorimeter (Parr Instruments Co. model 1356). Metabolizable energy intake was calculated by subtracting to the gross energy excretion in the form of feces, urine, and methane from the gross energy intake as previously reported (Arco-Pérez et al., 2017). Methane production was assumed as constant, representing 10.32% of the digestible energy (Aguilera, 2001). Urine was acidified with 1 M H_2SO_4 to keep pH < 2. Aliquots of feces and urine (2%) were stored at −20°C. The urinary concentration of purine derivatives and creatinine were determined an as indicator of the rumen microbial protein synthesis as previously described by Arco-Pérez et al., (2017). Finally, all animals (n = 10) were euthanized, gut was dissected, and the weight of the different gut compartments was recorded (reticulum-rumen, omasum, abomasum, small intestine, and large intestine). Rumen wall was dissected and samples (1 × 1 cm) from the cranial ventral and caudal ventral sacs were fixed with formaldehyde (4%) for histological examination. Tissues samples (5 × 5 mm) from the rumen wall were rinsed with PBS buffer and snap frozen for RNA extraction. To investigate the effects of the nutritional interventions on the rumen microbial fermentation, rumen content was filtered through a cheesecloth and solids were discarded given the small and variable proportion of solids in rumen samples. Then, rumen pH was measured and 5 subsamples of 0.8 mL were taken for ammonia (diluted with 0.2 mL of trichloroacetate 25 g/L), VFA (diluted with 0.8 mL of HCl 0.5 mol/L containing crotonic acid 0.8 g/: as internal standard and metaphophoric acid 200 g/L), protozoal optical counting (diluted in 1 mL of formaldehyde 75 mL/L and NaCl 9 g/L), lactate, and DNA extraction (snap frozen in liquid N) as previously reported (Belanche et al., 2019c). Cecum content was sampled, mixed 50:50 with PBS buffer, and filtered to determine concentration of ammonia and VFA to determine the effect of the dietary treatment on the hindgut fermentation.

Blood Parameters

After digestibility measurements (wk 22 of age), a blood sample from the jugular vein was obtained from each animal at 0900 h and placed in tubes with anticoagulant (K_3-EDTA). One blood subsample was centrifuged at 2,000 × g for 15 min at room temperature, and plasma was collected to determine plasma metabolites such as glucose, BHB, nonesterified fatty acids, cholesterol, urea, creatinine, total bilirubin, total proteins, albumin, globulins, aspartate aminotransferase, glutamate aminotransferase, alkaline phosphatase, and amylase. A second blood subsample was used to perform a blood count and to determine the concentration of the main blood cells morphotypes.

Ruminal Papillae Development and Gene Expression

Sample from rumen epithelium were cut in slices (1 mm wide) in triplicate and a photo was taken using optical stereomicroscope (M165FC Leica Microsystems). Measurements of the rumen papillae length and width were determined using the Leica Application Suite X Life Science software. Gene expression of targeted genes from the rumen papillae samples was assessed to determine the potential effect of the nutritional intervention on the rumen epithelial activity. Total RNA was extracted from 100 mg (fresh mater) of frozen epithelial samples from the rumen (cranial ventral sac). Samples were homogenized using a Bullet Blender homogenizer (Next Advance Inc.), and RNA was extracted using the TRIzol Reagent (Invitrogen) and purified using the RNeasy Mini-column (Qiagen) as previously reported (Abecia et al., 2017). The RNA concentration and quality was assessed using a ND-1000 spectrophotometer (NanoDrop Technologies Inc.) and a 2% agarose gel, respectively. Extracted RNA (1 μg) was reverse transcribed using the QuantiTect Reverse Transcription kit (Qiagen). This RNA was used as template for assessing the expression of genes related to (1) VFA absorption such as monocarboxylate transporter isoform 1 (*MCT1*) and putative transporter isoform 1 (*PAT1*), (2) VFA metabolism such as BHB dehydrogenase isoform 1 (*BHD1*) and 3-hydroxy-3-methylglutaryl-CoA lyase (*HMGCL*), (3) cell proliferation such as cyclin-D1, cyclin A, and cyclin-dependent kinase isoform 2 (*CDK2*), (4) epithelial growth such as insulin growth factor isoform 1 (*IGF1*) and IGF1 receptor (*IGF1R*), and (5) apoptosis such as B-cell lymphoma (*BCL-2*) and BCL-2 associated X protein (*BAX*) in the rumen samples. The expression of each gene was determined by quantitative PCR (**qPCR**) using SYBR Green (Sigma Aldrich) on a Bio-Rad iCycler (Bio-Rad Laboratories Ltd.). Primer sequences and qPCR annealing temperatures are described in Supplemental Table S2 (https://figshare.com/articles/journalcontribution/2023JDS_Suppldocx/22707937; Belanche et al., 2023). The relative expression of each gene was normalized to the β-actin as housekeeping gene, and the data were calculated by the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen, 2001).

Rumen Microbiota

To determine the effect of the dietary intervention on the concentration of the main microbial groups, rumen samples were freeze-dried and bead-beated for 2 rounds of 1 min using 2-mm beads. DNA was extracted using a commercial kit (QIAamp DNA Stool Mini Kit, Qiagen Ltd.). The DNA concentration was assessed using a NanoDrop spectrophotometer. Negative controls during the DNA extraction were included as quality controls. Rumen absolute abundance of total bacteria, methanogens, protozoa, and anaerobic fungi were determined by qPCR using specific primers (Supplemental Table S2) as previously described (Belanche et al., 2020). Cycling conditions were 95°C for 5 min, 40 cycles of 95°C for 15 s, 60°C for 30 s, and 72°C for 55 s, and 72°C for 1 min. The absolute amount of each microbial group, expressed as DNA copies per milligram of DM, was determined using serial dilutions of known amounts of standards. The qPCR standards consisted of the plasmid pCR4-TOPO (Invitrogen), with inserted 16S, mcrA, or 18S gene fragments for each microbial group. Rumen concentration of the main protozoal groups (Dehority, 1993) was determined using an optical microscope (Nikon Labophot).

A meta-taxonomic approach was conducted to determine the effects of the nutritional intervention on the rumen prokaryotic community as previously described (Palma-Hidalgo et al., 2021). Rumen DNA samples from CTL (n = 10) and PEY animals (n = 10) were sent to the IPBLN Genomic Facility (CSIC, Granada, Spain) for amplicon sequencing using the Illumina MiSeq V3 (600 cycles) kit (Illumina Inc.). A prokaryotic primer set (Supplemental Table S2) targeting the V3–V4 hypervariable region of the prokaryotic 16S rRNA gene which includes bacteria and methanogens, was used (Takahashi et al., 2014). Negative (water) and positive controls mock community (ZymoBIOMICS Microbial Community Standard, Zymo Research) were included as quality control. The PCR cycling conditions were 95°C for 3 min, 25 PCR cycles; 95°C for 30 s, 75°C for 10 s, 55°C for 30 s, 72°C for 30 s, and then 72°C for 5 min. Sequencing data were analyzed using QIIME2 (version 2021.4), first raw reads were primer-sorted, demultiplexed, and pair-ended. Low-quality reads (<Q30) were trimmed and chimeras were removed using chimera.vsearch (Edgar et al., 2011). All sequences were grouped into operational taxonomic units with a similarity cut-off of 97%. The resulting operational taxonomic units were taxonomically classified using the Silva 16S database (Silva_138 database). After removing singletons, the number of sequences per sample was normalized prior to statistical analyses. The relative abundance of each operational taxonomic unit along with the Good's coverage and α diversity indices (richness, Shannon's, Simpson's, and Chao1) were determined.

Long-Term Effects on Productivity

Dietary treatment was stopped at wk 22 of age and all animals joined the commercial herd. The persistency of the effects was assessed by monitoring the productive performance of the remaining female goats from the CTL (n = 24) and PEY group (n = 33). Female goats received natural mating and fertility, prolificacy, milk yield, and milk composition were monitored during the first lactation. Fertility was defined as the proportion of females that resulted pregnant after being assessed by ultrasound scan at d 42 postconception. Milk yield and milk composition was determined monthly (5 measurements per goat) using the information from the official milk control scheme as previously described (Belanche et al., 2019b). Milk samples treated with Azidiol (PanReac AppliChem) were analyzed for milk fat, protein, lactose, TS, and SCC by near infrared analysis, using the CombiScope FTIR 600 Dairy analyzer (Delta Instruments).

Statistical Analyses

Rumen fermentation parameters, nutrient use, blood metabolites, qPCR, and microbial diversity data were analyzed by 1-way ANOVA using the SPSS software (version 26.0, IBM Corp.) as follows:

$$Y_{ij} = \mu + T_i + e_{ij},$$

where Y_{ij} is the dependent, continuous variable, μ is the overall population of the mean, T_i is the fixed effect of the dietary treatment (CTL vs. PEY), and e_{ij} is the residual error. For the BW and ADG variables the sex effect (i.e., males vs. females) was included as a random factor. Data were inspected for normality using Shapiro-Wilk test and sequencing data and qPCR data and protozoal optical count data were $\log_{10}$ transformed before performing the ANOVA. The treatment effects on the rumen microbial communities were assessed based on the Bray-Curtis distance metrics using the unweighted pair group method with arithmetic mean function. Log_{10}-transformed sequencing data were analyzed by nonparametric PERMANOVA after 999 random unrestricted permutations of the raw data using PRIMER-6 software (PRIMER-E). The Monte Carlo test was used to prevent false positives. Principal coordinate analyses were conducted to illustrate

Table 1. Effect of dietary supplementation with a blend of plant extracts and yeast cell wall components (PEY), in relation to the control (CTL), on BW and ADG of goats in a commercial farm (n = 50)

Item	CTL	PEY	SEM	*P*-value
BW, kg				
Birth (d 1)	2.37	2.24	0.047	0.141
Weaning (wk 8)	8.00	8.60	0.158	0.019
Postweaning (wk 12)	9.98	10.9	0.197	0.004
ADG, g/d				
Preweaning (wk 1 to 8)	101	114	2.65	0.003
Postweaning (wk 8 to 12)	75.5	78.2	3.76	0.612

the dietary effects on the rumen microbial community. The Kruskal-Wallis nonparametric test was used to determine the dietary effects on the relative abundance of each bacteria taxa, rumen papillae gene expression, diarrheal score, and *E. coli* growth. For bacterial taxa, the *P*-values of their relative abundance were adjusted for multiple testing to decrease the false discovery rate (Benjamini and Hochberg, 1995).

RESULTS

Performance, Health, and Nutrient Utilization

Clay consumption progressively increased from birth until weaning (reaching 30 g/d) with no differences observed between treatments. Animals supplemented with PEY had a heavier BW than those in the CTL treatment at weaning (wk 8, $P = 0.019$) and at the postweaning period (wk 12, $P = 0.004$) on the commercial farm (Table 1). This resulted on a higher ADG during the preweaning ($P = 0.003$) but not during the postweaning period ($P > 0.1$).

Ten animals per treatment were moved to the research facilities at 19 wk of age to monitor feed utilization, diarrhea, and health-related metabolites in plasma (i.e., glucose, BHB, nonesterified fatty acid, cholesterol, urea, creatinine, total bilirubin, total proteins, albumin, globulins, aspartate aminotransferase, glutamate aminotransferase, alkaline phosphatase, amylase, and the main blood cells morphotypes). Within this subset, the PEY animals tended to have superior BW gain during the fattening period ($P = 0.078$, Table 2), resulting in heavier BW at wk 22 ($P = 0.043$). Control animals experienced a higher fecal score ($P = 0.004$) during the following week to transportation (Table 2), detecting *E. coli* as the most likely pathogen based on the fecal microbiological culture. This diarrheal process ceased after 1 wk of adaptation to the new environment and no major differences were noted on the blood concentration of health-related metabolites. Plant extract and YCW supplementation tended to increase blood nonesterified fatty acid ($P = 0.079$), urea ($P = 0.081$), albumin ($P = 0.024$), and alkaline phosphatase concentration ($P = 0.040$), whereas blood cell counts showed no differences between treatments (Supplemental Table S3; https://figshare.com/articles/journalcontribution/2023JDS_Suppldocx/22707937; Belanche et al., 2023).

Feed intake and digestibility was also monitored in a subset of animals per treatment (n = 10). Results indicated that animals supplemented with PEY tended to have higher concentrate feed intake at wk 21 ($P = 0.072$), but no differences were noted in forage intake, total DMI, gross energy intake, or ME intake between treatments (Table 2). The digestibility of the difference dietary nutrients was not affected by the treatments. Similarly, urinary excretion of purine derivatives, as indicator of the microbial protein synthesis, remained similar for both treatments when expressed in absolute terms (mmol/d) or in relation to the creatinine concentration.

Gut Development

The same subset of animals used for digestibility measurements was euthanized at wk 22 to assess the potential effects of the dietary intervention on the gastrointestinal tract (**GIT**) anatomical and physiological development (Table 3). Animals supplemented with PEY had heavier BW than CTL ($P = 0.013$), resulting in a higher weight of the full GIT ($P = 0.084$) and empty rumen ($P = 0.006$). These animals also tended to have a higher empty rumen weight when expressed as proportion of the GIT ($P = 0.067$), but not when expressed as proportion of the BW. Supplementation with PEY increased the rumen anatomical development by increasing the papillae length in the cranial ventral ($P = 0.014$) and caudal ventral sacs ($P = 0.002$), and the papilla width in the latter ($P = 0.051$). As a result, the papillae surface area was increased with PEY supplementation in the cranial ventral ($P = 0.096$) and caudal ventral sacs ($P = 0.003$). Plant extract supplementation increased expression of the *MTC1* gene ($P = 0.028$) in the rumen epithelium in comparison to the CTL treatment (Table 2). No significant differences were noted in the expression of other genes related to VFA absorption (*PAT1*, *BDH1*) and metabolism (*HMGCL*), rumen epi-

Table 2. Effect of dietary supplementation with a blend of plant extracts and yeast cell wall components (PEY), in relation to the control (CTL), on health indicators, blood parameters, feed intake, digestibility, and urinary excretion of purine derivatives in goats of 19 to 22 wk of age (n = 10)

Item[1]	CTL	PEY	SEM	*P*-value
Health indicator				
Fecal scoring,[2] wk 19	2.75	1.25	0.257	0.004
Escherichia coli,[3] wk 19	1.90	1.20	0.350	0.323
Fecal scoring,[2] wk 20	1.20	1.00	0.313	0.255
Performance				
BW at wk 22, kg	13.1	15.9	0.865	0.043
ADG wk 12 to 22, g/d	62.9	97.8	12.79	0.078
Blood parameter				
Glucose, mg/100 mL	78.0	76.3	3.720	0.828
BHB, mol/L	130	97.1	13.00	0.219
NEFA,[2] mol/L	238	444	58.00	0.079
Cholesterol, mg/100 mL	64.4	63.7	4.050	0.920
Urea, mg/dL	34.9	43.0	2.300	0.081
Creatinine, mg/100 mL	0.29	0.28	0.017	0.667
Total bilirubin, mg/dL	0.43	0.36	0.022	0.221
Total proteins, g/100 mL	6.09	5.96	0.153	0.731
Albumin, g/dL	2.25	2.59	0.070	0.024
Globulins, g/dL	3.84	3.36	0.158	0.168
Albumin/globulins	0.62	0.80	0.033	0.029
AST, U/L	81.4	80.8	22.80	0.947
ALT, U/L	18.6	21.6	5.150	0.228
ALP, U/L	138	255	30.40	0.040
Amylase, U/L	38.1	44.5	1.350	0.282
Intake				
Concentrate, g of DM/d	402	487	20.70	0.072
Oats hay, g of DM/d	105	98.5	12.80	0.797
Total DMI, g of DM/d	508	586	30.20	0.233
Gross energy, MJ/d	9.28	10.7	0.561	0.235
ME, MJ/d	5.86	6.31	0.334	0.536
Digestibility, %				
DM	71.2	67.5	1.36	0.207
OM	72.9	69.3	1.34	0.211
CP	70.4	65.8	1.48	0.148
NDF	56.6	52.0	1.86	0.260
ADF	49.1	47.4	1.84	0.688
Urinary excretion				
PD, mmol/d	3.63	3.13	0.525	0.514
Ratio PD/creatinine	1.99	2.52	0.476	0.372

[1]NEFA = nonesterified fatty acids; AST = aspartate aminotransferase, ALT = alanine transaminase; ALP = alkaline phosphatase; PD = purine derivatives.

[2]Fecal score from 1 (absence) to 4 (profuse diarrhea).

[3]*Escherichia coli* culture from 1 (absence of colonies) to 3 (massive colony growth).

thelial growth (*IGF1*, *IGF1R*), cell proliferation (cyclin A), and apoptosis (*BCL-2* and *BAX*).

Gut Fermentation and Microbiology

Dietary supplementation with PEY had a minor effect on the foregut and hindgut fermentation pattern (Table 4). No differences were detected on the rumen pH, ammonia-N, total VFA, and VFA molar proportions ($P > 0.05$). At the cecum, plant extract and YCW supplementation modified the fermentation pattern and PEY animals tended to have a higher butyrate ($P = 0.096$) and lower acetate molar proportion ($P = 0.056$) than the CTL animals. No differences were noted between treatments in terms of cecum pH, ammonia-N, and total VFA concentrations.

Quantitative PCR data showed that absolute abundance of total bacteria and methanogens in the rumen were unaffected by dietary supplementation with plant extracts and YCW components (Table 5); however, this supplementation significantly decreased the absolute abundance of anaerobic fungi ($P = 0.001$) and rumen protozoa ($P = 0.020$). These results were confirmed, to a certain extent, by optical microscopy, which also noted a tendency to lower rumen protozoal levels in the PEY animals ($P = 0.100$). The rumen protozoal community was dominated by the subfamily *Entodiniinae*, which representing over 93% of the population, had no differences between treatments.

Next generation sequencing yielded a small amount of methanogens sequences (<400 sequences per sample) that were considered insufficient and were, therefore,

Table 3. Effect of dietary supplementation with a blend of plant extracts and yeast cell wall components (PEY), in relation to the control (CTL), on the rumen anatomical development, papillae size, and rumen epithelial gene expression in goats of 22 wk of age (n = 10)

Item[1]	CTL	PEY	SEM	*P*-value
BW, kg	13.0	16.3	0.810	0.013
GIT, % BW	3.11	3.63	0.195	0.084
Empty rumen, kg	0.32	0.41	0.019	0.006
Empty rumen, % BW	2.53	2.52	0.101	0.964
Empty rumen, % GIT	10.3	11.5	0.404	0.067
Cranial ventral sac				
Papillae length, mm	3.75	4.83	0.229	0.014
Papillae width, mm	0.95	1.04	0.072	0.555
Papillae area, mm^2	8.33	11.5	1.221	0.096
Caudal ventral sac				
Papillae length, mm	2.84	4.06	0.218	0.002
Papillae width, mm	1.02	1.27	0.063	0.051
Papillae area, mm^2	6.99	12.0	0.992	0.003
Gene expression, fold change				
MCT1	1.00	2.31	0.443	0.028
PAT1	1.00	0.80	0.175	0.465
BDH1	1.00	0.72	0.164	0.361
HMGCL	1.00	2.25	0.479	0.291
Cyclin-D1	1.00	1.00	0.146	0.935
IGF1	1.00	1.05	0.558	0.950
IGF1R	1.00	2.01	0.766	0.935
Cyclin A	1.00	1.66	0.436	0.291
CDK2	1.00	1.55	0.289	0.291
BCL-2	1.00	1.66	0.350	0.808
BAX	1.00	2.22	0.438	0.062
BCL-2/BAX	1.06	0.80	0.156	0.167

[1]GIT = gastrointestinal tract.

discarded. On the contrary, sequencing yielded an average of 15,157 ± 2,298 (mean ± SD) raw bacterial sequences per sample, which, after quality filtering, resulted on an average of 13,732 ± 1,901 high quality sequences per sample. Sequencing depth was normalized the sample with the lowest sequencing depth (10,270 sequences per sample), providing a similar bacterial coverage for both treatments (average 0.84, Table 4). Sequencing reliability was conformed based on the positive (mock community) and negative control samples (data not shown). Animals fed the PEY diet had a lower bacterial richness ($P = 0.049$) than those in the CTL treatment, but no differences were noted in other diversity indexes (Shannon, Simpson, evenness, and Chao index). Principal coordinate analysis (Figure 1) showed a separation in the rumen bacterial community structure along the PCO1 according to the dietary treatment. These differences in the clustering pattern were confirmed by the PERMANOVA ($P = 0.027$). The meta-taxonomic analysis revealed that the *rumen* bacterial community was dominated by *Bacteroidota* (56.8%), *Firmicutes* (20.8%), Proteobacteria (12.9%), *Spirochaetota* (2.8%), *Fibrobacterota* (2.6%), and *Synergistota* (1.9%) with no differences in the major phyla between treatments (Supplemental Table S4; https://figshare.com/articles/journalcontribution/2023JDS_Suppldocx/22707937; Belanche et al., 2023). Moreover, PEY supplementation, in comparison to the CTL treatment, tended to decrease the relative abundance of the taxa *Oscillospiraceae*, *Prevotellaceae NK3B31 group*, *Clostridia UCG-014*, *Eubacterium ruminantium group*, *Ruminococcaceae UCG-002*, *Treponema*, *WCHB1–4*,

Table 4. Effect of dietary supplementation with a blend of plant extracts and yeast cell wall components (PEY), in relation to the control (CTL), on rumen and cecum fermentation patterns in goats of 22 wk of age (n = 10)

Item	CTL	PEY	SEM	*P*-value
Rumen fermentation				
pH	5.85	5.94	0.120	0.719
N-NH_3, mg/100 mL	14.2	19.7	2.750	0.187
Total VFA, m*M*	90.0	88.9	1.300	0.947
Acetate, %	62.5	61.4	0.700	0.782
Propionate, %	21.1	26.1	0.700	0.287
Butyrate, %	12.7	9.60	0.510	0.177
Isobutyrate, %	1.05	0.93	0.155	0.557
Isovalerate, %	1.26	1.03	0.220	0.571
Valerate, %	1.33	1.21	0.150	0.566
Cecum fermentation				
pH	6.34	6.40	0.050	0.594
N-NH_3, mg/100 mL	13.5	18.3	1.460	0.116
Total VFA, m*M*	84.8	80.3	1.200	0.722
Acetate, %	78.3	73.0	0.600	0.056
Propionate, %	13.9	15.2	0.400	0.291
Butyrate, %	5.34	8.89	0.510	0.096
Isobutyrate, %	0.87	0.80	0.169	0.770
Isovalerate, %	0.59	0.83	0.187	0.405
Valerate, %	0.98	1.28	0.160	0.154

Table 5. Effect of dietary supplementation with a blend of plant extracts and yeast cell wall components (PEY), in relation to the control (CTL), on the rumen concentration of the main microbial groups and on the bacterial diversity and taxa relative abundance (%) in goats of 22 wk of age (n = 10)[1]

Item[2]	CTL	PEY	SEM	*P*-value
Microbes, $\log_{10}$ copies/mL				
Bacteria	12.2	11.9	0.160	0.227
Methanogens	6.23	6.63	0.271	0.304
Protozoa	7.27	6.41	0.247	0.021
Anaerobic fungi	5.49	3.99	0.281	0.001
Optical counting				
Total protozoa, log cell/mL	6.15	5.29	0.290	0.140
Subf. *Entodiniinae*, %	93.5	94.2	1.105	0.762
Subf. *Diplodiniinae*, %	6.53	5.83	1.105	0.762
Bacterial diversity				
Richness, OTU[2]	517	405	37.4	0.049
Shannon	4.71	4.44	0.165	0.267
Simpson	0.97	0.96	0.008	0.396
Evenness	0.76	0.74	0.018	0.621
Chao1	602	478	61.00	0.172
Good's coverage	0.84	0.83	0.029	0.794
Relative abundance,[3] %				
p_Actinobacteriota	0.03	1.08	0.486	0.099
f_Erysipelotrichaceae	0.05	0.19	0.048	0.099
f_Oscillospiraceae	2.79	0.59	0.640	0.055
g_Prevotellaceae_NK3B31_group	0.03	0.03	0.014	0.047
g_Prevotellaceae_UCG-004	0.14	0.00	0.040	0.087
g_Bacteroidetes_BD2–2	0.02	0.00	0.008	0.091
g_Desulfobulbus	0.01	0.07	0.017	0.040
g_Clostridia_UCG-014	0.43	0.10	0.072	0.042
g_Erysipelotrichaceae_UCG-009	0.00	0.04	0.009	0.055
g_Eubacterium_ruminantium	0.17	0.03	0.048	0.031
g_Papillibacter	0.05	0.00	0.018	0.091
g_Ruminococcaceae	0.06	0.01	0.015	0.076
g_Schwartzia	0.08	0.00	0.027	0.091
g_UCG-002	1.10	0.34	0.242	0.094
g_Eubacterium_hallii_group	0.01	0.00	0.002	0.091
g_Erysipelotrichaceae_UCG-008	0.00	0.04	0.017	0.091
g_Absconditabacteriales_(SR1)	0.17	0.00	0.054	0.053
g_Treponema	3.93	1.58	0.784	0.099
g_WCHB1–41	0.16	0.08	0.033	0.039
s_Fibrobacter_succinogenes	0.42	0.19	0.093	0.081
s_Selenomonas_ruminantium	0.26	1.24	0.265	0.051

[1]Only taxa that tended to differ ($P < 0.1$) between treatments are shown.

[2]Subf. = subfamily; OTU = operational taxonomical unit.

[3]Adjusted *P*-values to minimize the false discovery rate. p_ = phylum, f_ = family, g_ = genus, s_ = species.

and *Fibrobacter succinogenes* and led to undetectable levels of *Prevotellaceae UCG-004*, *Bacteroidetes BD2–2*, *Papillibacter*, *Schwartzia*, *Eubacterium hallii group*, and *Absconditabacteriales SR1* ($P < 0.1$). On the contrary, PEY animals tended to have higher rumen relative abundances for the taxa *Actinobacteriota*, *Erysipelotrichaceae*, *Desulfobulbus*, *Erysipelotrichaceae UCG-099*, and *Selenomonas ruminantium* ($P < 0.1$).

Long-Term Effects on Reproduction and Milk Yield

A total of 57 female goats were monitored during the first lactation to investigate the persistency of the effects of the early-life nutritional intervention on a commercial farm (Table 6). A higher fertility rate was observed in the PEY (64.7%) than in the CTL group (54.2%) even though all of them have similar first partum age and prolificacy. No differences were observed between treatments in terms of milk yield, milk components (fat, protein, and lactose), and SCC.

DISCUSSION

Effects on Feed Utilization and Health

This study evaluated the effects of supplementing a novel plant extract blend containing turmeric, thymol, and YCW components as the main active components. Turmeric is rich in curcumin, which is known to possess chemo-preventive properties due to its antiinflammatory, antioxidative, and antimicrobial activities, making it useful to prevent arthritis, mastitis, and GIT infec-

Table 6. Long-term effects of dietary supplementation in early life with a blend of plant extracts and yeast cell wall components (PEY), in relation to the control (CTL), on the reproductive indicators and milk yield in dairy goats during the first lactation

Item	CTL	PEY	SEM	*P*-value
Number of animals	24	34		
First partum age, mo	15.1	15.3	0.155	0.571
Fertility, %	54.2	64.7		
Prolificacy, kids/doe	1.08	1.05	0.052	0.665
Milk production				
Milk yield, kg/d	2.19	2.25	0.090	0.720
FPCM,[1] kg/d	2.48	2.55	0.099	0.693
Milk component				
Fat, %	4.99	5.02	0.117	0.879
Protein, %	3.47	3.51	0.050	0.628
Lactose, %	4.98	4.98	0.031	0.926
SCC, $\log_{10}$ cells/mL	5.45	5.42	0.080	0.675

[1]FPCM = fat- and protein-corrected milk calculated for 4% fat and 3.3% protein content.

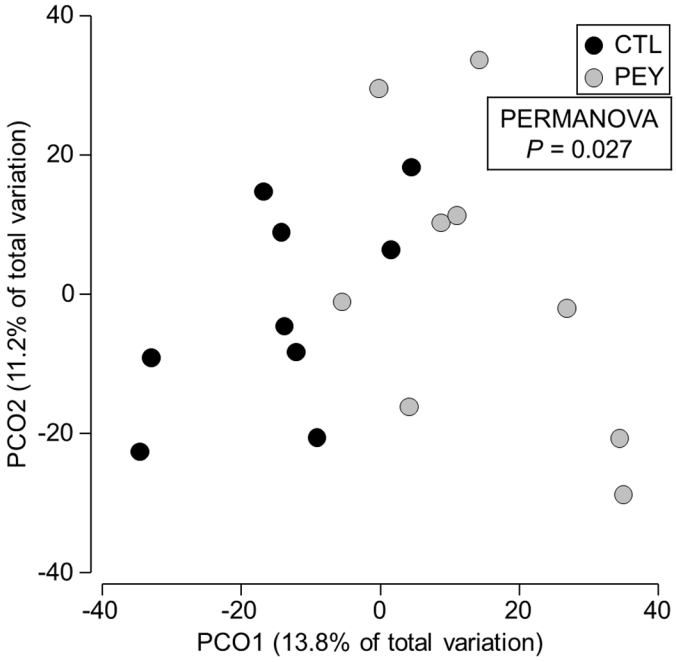


Figure 1. Principal coordinate (PCO) analysis of the rumen bacteria community, illustrating the effect of dietary supplementation with a blend of plant extracts and yeast cell wall components (PEY), or absence of supplementation (CTL) in goats. The PERMANOVA value is provided based on the Bray-Curtis dissimilarity.

tions (Lans et al., 2007). As a result, administration of curcumin has been shown to use the antiinflammatory response pathway to protect the intestine against bacterial invasion, leading to a significant reduction in the GIT inflammation and oxidative stress (Cho and Park, 2015). This mode of action could explain the lower incidence of diarrheal events (fecal score) observed in PEY than in CTL kids during the subsequent days after the stress caused by the transportation and the new environment (wk 19). These diarrheal events led to lower BW gain of the CTL lambs during the fattening period (−36%), although they ceased after 1 wk and no differences were noted in health-related indicators at wk 22. The presence of a compensatory growth of the CTL animals during the postweaning period (wk 8 to 12) could explain the lack of differences in the BW gain during this period, and the positive effects on the animal health and performance only re-appeared when the immune system was under stressing situations as previously reported (Cervantes-Valencia et al., 2016). In addition to health benefits, a recent dose-response study over a 30-d period reported that curcumin dietary supplementation (300 mg/kg of concentrate) in lambs increased BW (+5.7%) and BW gain (+40%), leading to leaner carcasses with higher PUFA content in lamb meat (Marcon et al., 2020). This observation could be due to changes in the lipid metabolism and antioxidant properties as described in quail (Marchiori et al., 2019). The use of a much lower dose of curcumin (0.91 mg/kg) in our study agrees with these findings because PEY animals had higher BW gain during the preweaning period (+12.9%) resulting on heavier BW at weaning (+7.5%) and at 12 wk of age (+9.2%). According to Molosse et al. (2019), the enhancement of BW and BW gain by curcumin supplementation was linked to stimulation of serum creatine kinase activity. This enzyme adds phosphate groups to creatine in the muscle cells and is thought to improve the energy metabolism accumulating phosphocreatine that represents a temporal energy buffer to prevent falls in global adenosine triphosphate content. Jaguezeski et al. (2018) also reported positive effects of supplementing curcumin on NDF digestibility, milk production, and milk protein content in dairy ewes. A higher N retention was also reported in beef cattle supplemented with curcumin as a result of lower rumen ammonia concentration and fecal N excretion (Vorlaphim et al., 2011). Some of these effects were not evident in our study because similar gross energy intake and feed digestibility were observed between both treatments, suggesting that the multifactorial efficacy of curcumin consumption may be modulated by the type of diet, livestock species, or age of the animals, or all these factors.

Thymol supplementation in dairy cows has been linked to a higher feed efficiency (Hristov et al., 2013), but no differences in feed intake and nutrient digestibility have been related to thyme oil supplementation in beef (Khorrami et al., 2015) and dairy cattle (Benchaar, 2021). In our study, DMI was not measured during the entire experimental period, but it was monitored in a subset of animals during 1 wk prior to the digestibility assessment. For these animals, no differences were observed in the total DMI, but PEY animals tended

to have a higher (+21.1%) concentrate feed intake than CTL animals, possibly as a result of a better feed palatability. This increased concentrate feed intake can justify the numerical increase in the rumen propionate proportion (+23.7%) in PEY animals, and the higher BW gain during the fattening period. This observation could indicate a higher efficiency of energy utilization in PEY animals.

The effects of EO supplementation on the total VFA concentrations has been described as highly variable depending on the type of EO, dose, rumen pH, and type of diet (Calsamiglia et al., 2007). In our study, no differences were noted in the rumen VFA concentrations between treatments. Similar findings were observed in dairy cows supplemented with oregano (Hristov et al., 2013). Rumen VFA concentration depends not only on the VFA production but also on its absorption rate through the rumen epithelium. In this sense, a recent study (Zhang et al., 2021) demonstrated that supplementing the diet of dairy cows with up to 130 mg/d of oregano EO (containing thymol) increased the rumen digestive ability by modulating epithelial development (papillae length), rumen enzymatic activity (increased β-glucosidase and cellulase in detriment of amylase), and concentration of fermentation products (propionate and butyrate in detriment of acetate). These observations were partially confirmed in our study because PEY animals had substantially longer papillae in the cranial ventral (+28.8%) and caudal ventral sac (+42.9%) in comparison to CTL animals. This greater papillae development implied an increased absorptive surface area (+38.0% and +71.7%, respectively), which along with the higher expression of genes related to VFA absorption (*MCT1*) could result in a higher VFA absorption through the rumen epithelium. Although no differences between treatments were detected for the expression of targeted genes related to VFA absorption (*PAT1*), VFA metabolism (*BHD1* and *HMGCL*), cell proliferation (cyclin-D1, cyclin A, and *CDK2*), epithelial growth (*IGF1* and *IGF1R*), and apoptosis (*BCL-2* and *BAX*), many other genes are involved in nutrient absorption and rumen epithelial development, which warrants further study. Regardless, rumen epithelial development is a complicated process and several factors such as the DMI, rumen pH, fermentation pattern, or rumen microbiome can affect the gene expression in the ruminal epithelium (Baldwin and Connor, 2017). The improved rumen anatomical development noted in PEY animals, in comparison with CTL animals, was also supported by a higher empty rumen size when expressed in absolute (+27.3%) and relative terms in relation to the total GIT weight (+11.7%). The reason for the increased rumen size is unknown, but it could be linked to a higher solid feed intake (and BW gain) and changes in the rumen microbiota early in life as previously suggested (Diao et al., 2019). Moreover, a higher rumen absorptive area could imply a decrease in the rumen VFA concentration leading to a higher VFA absorption and blood VFA concentrations as previously reported in calves fed an EO blend containing thymol and prebiotics (Liu et al., 2020). Using an in vitro approach, Cardozo et al. (2005) reported that the effects of thymol supplementation is pH dependent, promoting an increase in the acetate-to-propionate ratio with high-forage diets (pH 6.5) and a decrease of this ratio with high-concentrate diets (pH 5.5). Several authors have also reported a decrease in the acetate-to-propionate ratio in beef cattle supplemented with thyme oil (Khorrami et al., 2015) and thymol (Zhang et al., 2021). This shift in the rumen fermentation could partially explain the higher energy efficiency in the feed observed as a result of thymol supplementation in ruminants fed high-concentrate diets and low rumen pH (Liu et al., 2020). However, our animals had an intermediate rumen pH (average 5.90), which could explain the lack of clear effects of the PEY diet on the rumen fermentation. Further research is needed to evaluate the effects of this plant extract blend on more extreme dietary conditions.

The PEY diet also contained YCW oligosaccharides such as mannans and β-glucans. It has been reported that these oligosaccharides can improve the GIT histological development, absorb pathogens, and enhance the release of cytokines, leading to improved immunity in broilers (Liu et al., 2018). In ruminants, several studies have demonstrated that YCW supplementation is beneficial for milk production (Nocek et al., 2011) and ruminal development (Lesmeister et al., 2004). In addition, Terré et al., (2007) reported that mannan oligosaccharides from YCW stimulated the starter intake (+13.4%) during the postweaning period in calves as observed in the present study. Moreover, in line with our observations, recent studies in calves (Xiao et al., 2016; Ma et al., 2020a) noted that dietary supplementation with YCW can led to enhanced rumen papillae development in terms of length, width, and surface area. These studies also described positive effects on villus height-to-crypt depth ratio in the jejunum and superior intestinal barrier function, aspects that were not measured in our study. Hindgut fermentation is intimately linked with the gut health and is determined by the availability of fermentable substrate and by type of gut microbiota (Guilloteau et al., 2010). Although, no substantial differences were observed on the cecum fermentation parameters, it was noted that PEY supplementation tended to increase the butyrate proportion (+66.5%) in detriment to acetate (−6.8%). Butyrate is preferentially taken up by the gut epithelial

cells where it is actively metabolized to produce energy; therefore, increased butyrate level is often considered as an indicator of gut health (Guilloteau et al., 2010). Further analysis could help to understand whether these fermentative differences are linked to changes in the hindgut microbiota.

Modulation of the Rumen Microbiota

Previous studies have shown beneficial effects of in-feed administration of turmeric in lambs to control coccidiosis and to limit the host inflammatory reaction, resulting in increased BW gains (Cervantes-Valencia et al., 2016). In our study, PEY supplementation promoted a modulation of the rumen bacterial community promoting subtle changes of minor taxa. Moreover, an antiprotozoal activity was also observed because lower rumen protozoal levels were detected when measured by qPCR and by optical microscopy. A similar antiprotozoal effect has been described in beef cattle fed concentrates containing up to 0.2% added curcumin (Vorlaphim et al., 2011). A decrease of the protozoal genera *Isotricha*, *Diploplastron-Eremoplastron*, *Polyplastron*, and *Ophryoscolex* have also been described when thymol was supplemented in vitro at 400 mg/L (Yu et al., 2020). The mechanism by which PEY decreased rumen protozoa is unknown, but it has been described that the thymol antiprotozoal (Tasdemir et al., 2019) and antifungal activity (Marchese et al., 2016) could be ascribed to an increased membrane ion permeability causing the cell death. A meta-analysis showed that absence of the rumen protozoa has positive effects on the N metabolism and animal growth as a result lower bacterial protein breakdown by protozoa (Newbold et al., 2015). However, these beneficial effects were not detected in the present study because both experimental treatments showed similar rumen ammonia concentration and urinary excretion of purine derivatives as indicators of the rumen protein breakdown and microbial protein flow to the small intestine, respectively. On the contrary, our study indicated that the turmeric and thymol antimicrobial effects could be extrapolated to other eukaryotic microbes such as anaerobic fungi. It has been described that curcumin may act as a potent fungicide due to a downregulation of $\Delta^{5,6}$ desaturase (ERG3), leading to significant reduction in ergosterol and ultimately the death of the fungal cell via generation of reactive oxygen species (Sharma et al., 2010). These observations are in line with an in vitro study in which a linear decrease in the concentration of protozoa, anaerobic fungi, and bacteria was noted when supplemented with increasing turmeric doses (Aderinboye and Olanipekun, 2021). In previous studies, we have observed that low absolute abundances of rumen protozoa and anaerobic fungi represent a microbial adaptation for efficient utilization of high-starch diets as used in the present study (Belanche et al., 2012, 2019a). Sulistyowati et al. (2014) identified some additive effects when turmeric was supplemented in combination with yeast, leading to greater reductions in the protozoal numbers and in vitro feed digestibility than when each additive was supplemented alone. The antiprotozoal effect was observed in our study, but without having a negative effect on feed digestibility, possibly as a compensatory mechanism induced by changes in the rumen bacterial community. Similar absence of effects on feed digestibility has been reported in lambs supplemented with turmeric powder (Khalesizadeh et al., 2011).

Tyagi et al. (2015), after using fluorescent probes, demonstrated that curcumin exerts a broad bactericidal activity by damaging the bacterial membrane of gram-positive (*Staphylococcus aureus* and *Enterococcus faecalis*) and gram-negative potential pathogens (*E. coli* and *Pseudomonas aeruginosa*). Similarly, a direct binding has been described between YCW oligosaccharides and potential pathogenic bacteria (*E. coli*, *Salmonella*, and *Listeria*) inhibiting their proliferation in the GIT and helping to prevent subsequent infections (Broadway et al., 2015), which may be linked with the lower abundance of *E. coli* in feces noted in the present study (−55%). Moreover, PEY animals had higher levels of *Erysipelotrichaceae* and *Desulfobulbus*, which have been associated with lower incidence of diarrhea (Ma et al., 2020b) and early starter feed consumption in young ruminants, respectively (Liu et al., 2017). These microbial differences could be associated with a lower susceptibility of the PEY animals to experience diarrheal events after stress.

Borchers (1965) was the first to report the potential benefit of EO on rumen fermentation when observing that the addition of thymol to rumen fluid in vitro resulted in the accumulation of aminoacidic N and the reduction of ammonia-N concentrations, suggesting that thymol inhibited deamination. This hypothesis was not confirmed in our study because rumen ammonia-N concentration was similar between treatments. Due to the presence of a hydroxyl group, compounds with phenolic structures such as thymol, are more effective as antimicrobials in comparison with other nonphenolic secondary plant metabolites (Calsamiglia et al., 2007). Moreover, the small molecular weight of thymol allows it to cross the cell wall and gain access to the bacterial membrane, causing conformational changes, increasing permeability, and exerting a wide-spectrum antimicrobial activity against gram-positive and gram-negative bacteria (Calsamiglia et al., 2007; Zhang et al., 2021). However, the narrow margin of security between optimal

and toxic dosage observed for thymol often make that the expected effects are not always in the desired direction (Castillejos et al., 2006). In a recent dose-response study, it was reported that thymol supplementation decreased the bacterial diversity when rumen microbiota from goats was incubated in vitro (Yu et al., 2020). In our study, PEY supplementation also decreased the bacterial richness without further changes in other diversity indicators (Shannon and Simpson index). This observation seems to imply a modulation of the rumen bacterial community structure, as revealed by the PERMANOVA analysis, consisting of the disappearance of some low abundant taxa (i.e., *Prevotellaceae_UCG-004*, *Bacteroidetes_BD2–2*, *Papillibacter*, *Schwartzia*, *Eubacterium_hallii_group*, *Absconditabacteriales_SR1*) along with a greater homogeneity in the relative abundance of the remaining taxa. Moreover, PEY supplementation also decreased the relative abundance of *Papillibacter*, *Erysypelotrichaceae*, *Ruminococcadeae*, and *Treponema*, which have been described as indicators of low-feed efficiency in steers (Lopes et al., 2021). On the contrary, PEY animals showed an adaptation to the higher concentrate intake, characterized by higher relative abundance of *Selenomonas ruminantium*, which uses starch and simple sugars as the main energy source, and lower levels of fibrolytic taxa (i.e., *Fibrobacter succinogenes* and *Eubacterium ruminantium*), which mostly uses cellulose, hemicellulose, and cellobiose.

Yu et al., (2020) showed a negative in vitro effect of thymol on the relative abundance of certain rumen methanogens (i.e., *Metanobrevibacter* and *Methanomassiliicoccaceae*). Our study showed low and similar relative abundance of methanogens between treatments. These low levels of rumen methanogens could be due to fact that the experimental animals were artificially reared in the absence of contact with adult animals as previously shown (Belanche et al., 2019c). Moreover, the fact that rumen samples were only collected at 22 wk of age made difficult to speculate about the potential effects of this nutritional intervention on the rumen microbial colonization process. Thus, further longitudinal studies covering earlier stages of development (from birth to adulthood) are needed to truly assess the effects of PEY supplementation on the rumen anatomical and microbial development.

Long-Term Effect on Reproduction and Milk Yield

This study demonstrated that the rumen microbial and functional development can be modulated by early-life nutritional interventions based on the use of bioactive compounds; therefore, it was hypothesized whether this intervention could have positive long-lasting effects on the animal performance later in life (Yáñez-Ruiz et al., 2015). Our findings suggested that the supplementation with plant extract and YCW components, despite stopping at 5 mo of age, had positive implications during the first-conception age (average 10.2 ± 0.73 mo of age), resulting in a higher fertility rate (+19.4%). Higher gonadal activity and development has been described in Hu rams supplemented with up to 900 mg of curcumin per sheep (Jiang et al., 2019). In our study, the potentially superior ovary development or activity, or both, could also be indirectly associated with an increased BW gain in PEY animals (Robinson et al., 2006). On the contrary, no differences were noted in milk yield and milk components between treatments, suggesting that the higher rumen anatomical or microbiological development, or both, observed in PEY animals tended to disappear over time when the nutritional intervention ceased. This lack of persistency of the effects could be due to a compensatory rumen anatomical development in the CTL animals later in life and to the high redundancy among rumen microbes which occupy similar niches and metabolic pathways (Weimer, 2015).

CONCLUSIONS

This study demonstrated that the supplementation of young goats with a plant extract blend (PEY) containing turmeric, thymol, and YCW oligosaccharides favored the rumen anatomical and papillae development. This supplementation increased concentrate feed intake and decreased the rumen bacterial diversity and absolute abundances of protozoa and anaerobic fungi, as an adaptation to better digest concentrate feeds. As a result of a higher nutrient uptake, supplemented animals had increased BW gain in early life and higher fertility rates later in life. However, this study was unable to associate these positive changes in the rumen anatomical and microbiological development with changes in the rumen fermentation pattern, feed digestibility, or milk production. These findings suggest that early life supplementation with PEY could be considered as a sustainable nutritional strategy to increase BW gain and to optimize the rumen anatomical and microbiological in young ruminants, despite having minor productive implications later in life.

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